Group C · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**49**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

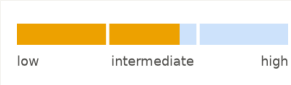
Sample: **TCGA-A2-A3XY-01** · Aneuploidy score: **8** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0.02** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

60th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 20%

Model-estimated progression time

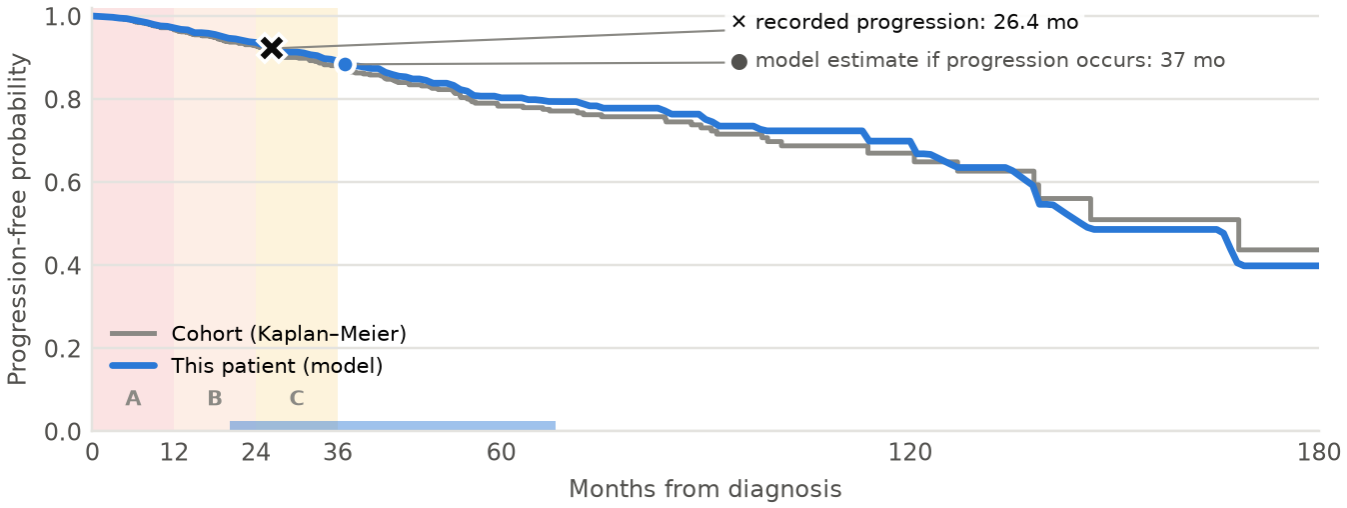
≈ 37 months

Window > 36 months · IQR 20–68 mo if progression occurs

Recorded progression

26.4 months

Group C · Progression in year 3
✗ recorded window differs · |error| 11 mo



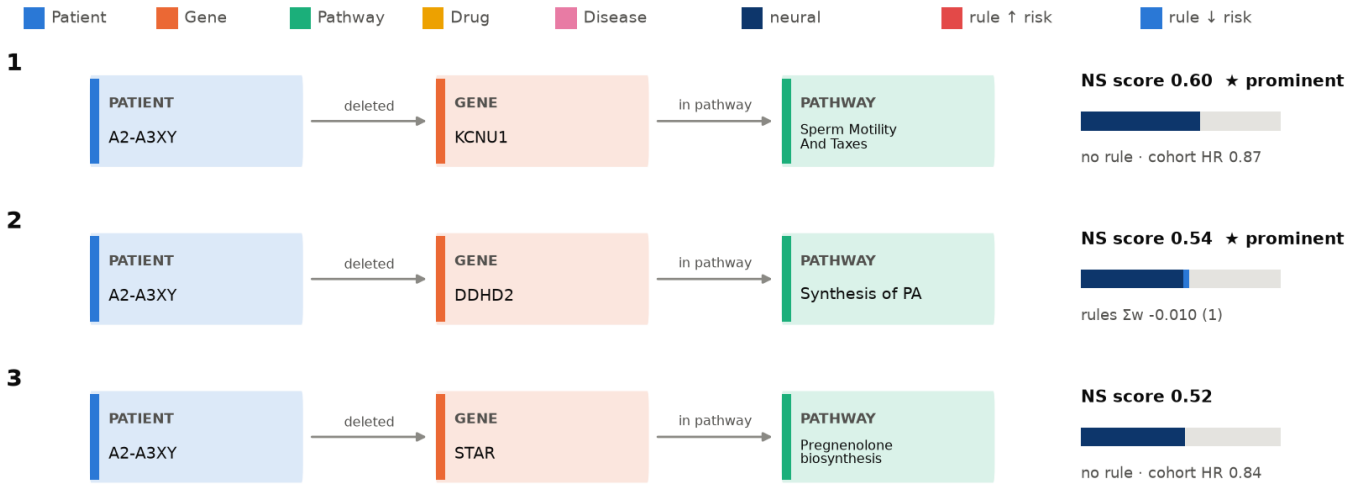
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 49-year-old patient has a subtype-unassigned tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **60th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Stage(p, II)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **KCNU1** to **Sperm Motility And Taxes**. Progression was recorded at 26.4 months; the model's estimate given progression was 37 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

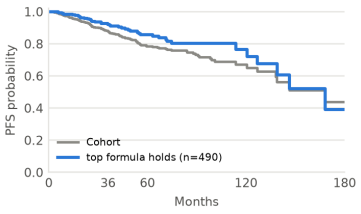


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A2-A3XY** deleted → **KCNU1** in pathway → **Sperm Motility And Taxes**	99	No learned formula fires on this path; its score is neural only. cohort: HR 0.87, n = 97 carriers	**0.60**
2 ★	**TCGA-A2-A3XY** deleted → **DDHD2** in pathway → **Synthesis of PA**	85	↓ **slower progression** · Σw -0.010 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) cohort: HR 0.80, n = 189 carriers	**0.54**
3	**TCGA-A2-A3XY** deleted → **STAR** in pathway → **Pregnenolone biosynthesis**	87	No learned formula fires on this path; its score is neural only. cohort: HR 0.84, n = 196 carriers	**0.52**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

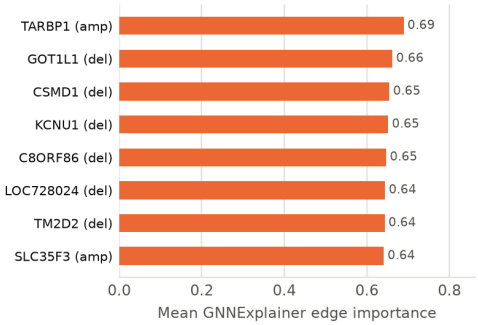
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.041 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	337 pts
-0.010 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) [KG]	single atom	187 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	Homozygous deletion	GISTIC ±2	43	0.32
PIK3CA · driver	High-level amplification	GISTIC ±2	70	0.31
TARBP1	High-level amplification	GISTIC ±2	0	0.69
GOT1L1	Homozygous deletion	GISTIC ±2	0	0.66
CSMD1	Homozygous deletion	GISTIC ±2	0	0.65
KCNU1	Homozygous deletion	GISTIC ±2	1	0.65
C8ORF86	Homozygous deletion	GISTIC ±2	0	0.65
LOC728024	Homozygous deletion	GISTIC ±2	0	0.64
TM2D2	Homozygous deletion	GISTIC ±2	0	0.64
SLC35F3	High-level amplification	GISTIC ±2	0	0.64
LETM2	Homozygous deletion	GISTIC ±2	0	0.63
PROSC	Homozygous deletion	GISTIC ±2	0	0.63



39 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 52 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A1ES	35	II	missing	0.80	progressed at 113.8 mo (group D)
TCGA-E2-A1IE	61	II	LumA	0.75	progressed at 59.5 mo (group D)
TCGA-WT-AB41	55	II	LumA	0.74	no progression, followed 53.0 mo
TCGA-B6-A0WT	61	II	LumA	0.73	progressed at 127.0 mo (group D)
TCGA-BH-A204	80	II	LumB	0.73	no progression, followed 83.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.